

ALGEBRA GENERAL EXAM JANUARY 10, 2004

Problem 1

(10 point) Let $G = \{g_1, \dots, g_n\}$ a finite abelian group. Show that the product $P := g_1 \cdot \dots \cdot g_n$ is of order 1 or 2 .

Problem 2

(10 point) Let H be the subgroup of $G := \mathbb{Z} \oplus \mathbb{Z} = \{(m, n) \mid m, n \in \mathbb{Z}\}$ generated by $(1, 2)$ and $(3, 4)$. Identify the quotient group G/H .

Problem 3

(15 point)

(a) If $A \in M_n(\mathbb{R})$ is idempotent, i.e. $A^2 = A$, then it is diagonalizable.

(b) Two idempotent matrices $A, B \in M_n(\mathbb{R})$ are similar if and only if they have the same rank.

Problem 4

(10 point) Show that a principal ideal in $\mathbb{Z}[x]$ can never be maximal.

Problem 5

(15 point) Let ξ be a primitive 9 -th root of unity. Let $K = \mathbb{Q}(\xi)$ and $F = \mathbb{Q}(\xi + \xi^{-1})$.

(a) Show that $[K : F] = 2$.

(b) Show that the extension $F/\mathbb{Q}$ is normal.

Problem 6

Let V and W be finite dimensional vector spaces over a field k , and let $f : V \rightarrow V$ and $g : W \rightarrow W$ be linear operators. One can define a linear operator $f \otimes g : V \otimes_k W \rightarrow V \otimes_k W$. Show that $\text{Tr}(f \otimes g) = \text{Tr } f \cdot \text{Tr } g$.

Problem 7

(15 point) Let S be a finite set acted upon by a finite group G . Denote by $\mathbb{C}(S)$ the complex vector space of complex-valued functions on S .

(a) Show that the map $G \times \mathbb{C}(S) \rightarrow \mathbb{C}(S), (x, f) \mapsto x.f$, defines a G -action on $\mathbb{C}(S)$. Here

$x.f \in \mathbb{C}(S)$ is defined by $x.f(s) = f(x^{-1}s)$ for all $s \in S$.

(b) Show that the dimension of the subspace of G -fixed points in $\mathbb{C}(S)$ is equal to the number of G -orbits in S .

Problem 8

(15 point) TRUE-FALSE. You do not need to justify your answers.

(a) Every matrix in $SL_n(\mathbb{R})$ has a positive real eigenvalue if n is odd.

(b) The Galois group of $x^3 + 1$ is isomorphic to S_3 .

(c) A group of order 99 must have a unique normal subgroup of order 11.

(d) There exists a unique finite field of order 6 up to isomorphism.

(e) $\mathbb{Z}[x]$ is a Euclidean domain.